

Service Publication

Diesel Engine
16V 396 TE84

Description and
Operation Manual
M011050/00E

DETROIT DIESEL



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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents which lists the Groups and Subgroups.

The Groups in Sections B and G correspond to the Groups in the Spare Parts Catalog and other Product Support publications.

Using the Table of Contents to trace an entry

- ➡ Table of Contents (general)
- ➡ Table of Contents (section)
- ➡ Group / Subgroup
- ➡ Heading

Using the Alphabetical Index to trace an entry

- ➡ Term in Alphabetical Index
- ➡ Group / Subgroup
- ➡ Heading

Publication Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

A Product Summary
Engine summary, technical data

B Description
System and subassembly descriptions

C Operating Instructions
Task descriptions

D Maintenance Schedule
Schedule and maintenance frequency chart

E Troubleshooting
Fault conditions, probable causes, remedies

F Drawings
Engine cross-sections

G Task Description
Inspection and maintenance

K Preservation
Engine preservation

Z Appendix
Supplementary documentation

? Index
Alphabetical index of key terms

Amendment Service!

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M011050/00E

**Description and
Operation Manual**

16V 396 TE 84



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MTU Friedrichshafen GmbH
Department SCT
88040 Friedrichshafen
GERMANY

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Diesel EnginePart A
Product Summary

The product summary gives details of the engine and contains technical data such as engine power, general engine specifications, engine weight, oil and coolant capacities, settings and operational data.

ENGINE LAYOUTKey to Engine Model Designation 16 V 396 TE 84

- 16 = number of cylinders
V = Vee configuration
396 = engine series - 100 times displacement of one cylinder in liters
T = exhaust turbocharging
E = charge air-to-coolant cooling in split-circuit coolant system
8 = engine for marine propulsion
4 = design digit

Engine Power

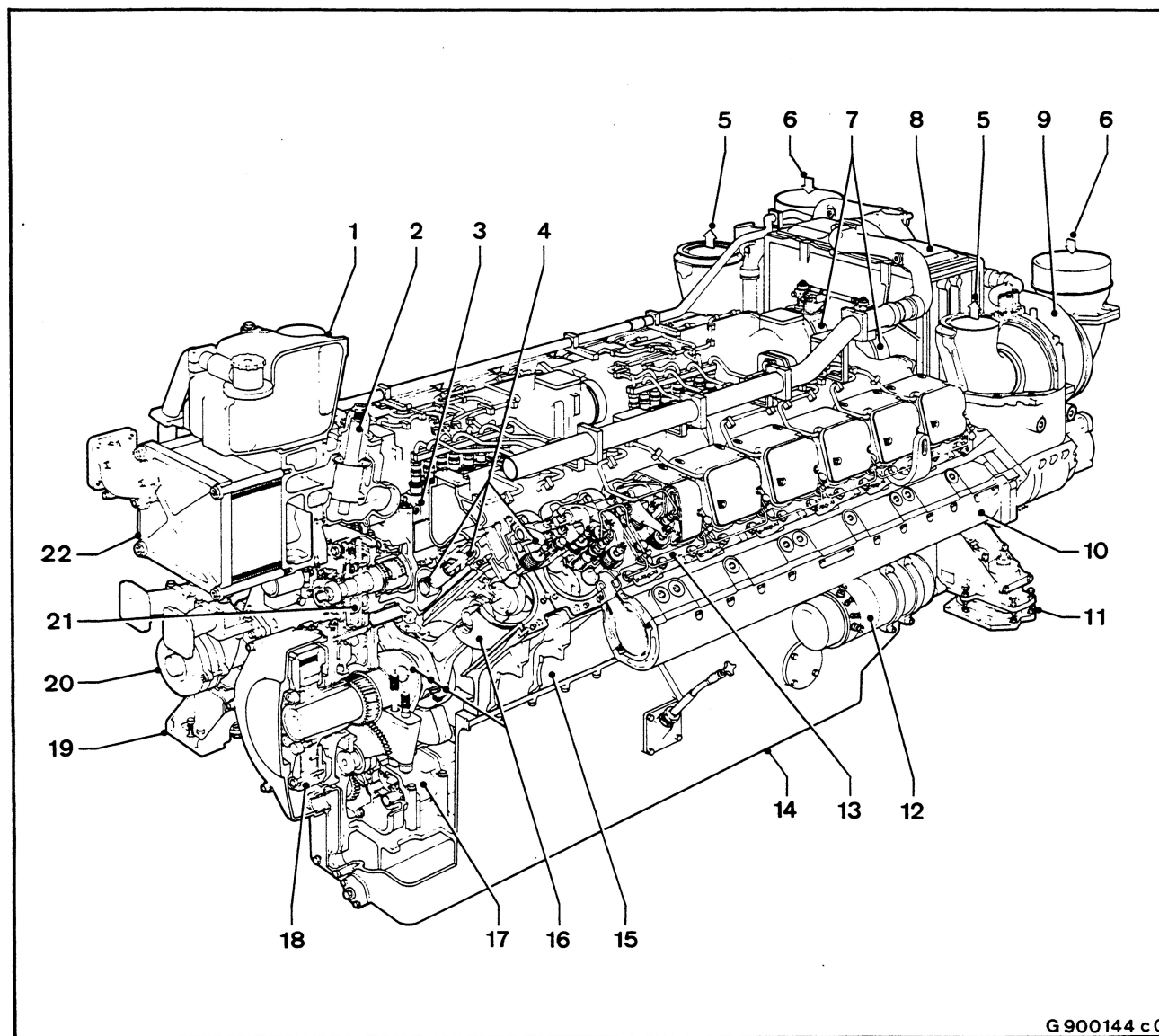
Continuous power (fuel stop power) 2000 kW at 1900 rpm

Reference conditions

- intake air temperature 25°C
- raw water temperature 32°C
- barometric pressure 1000 mbar



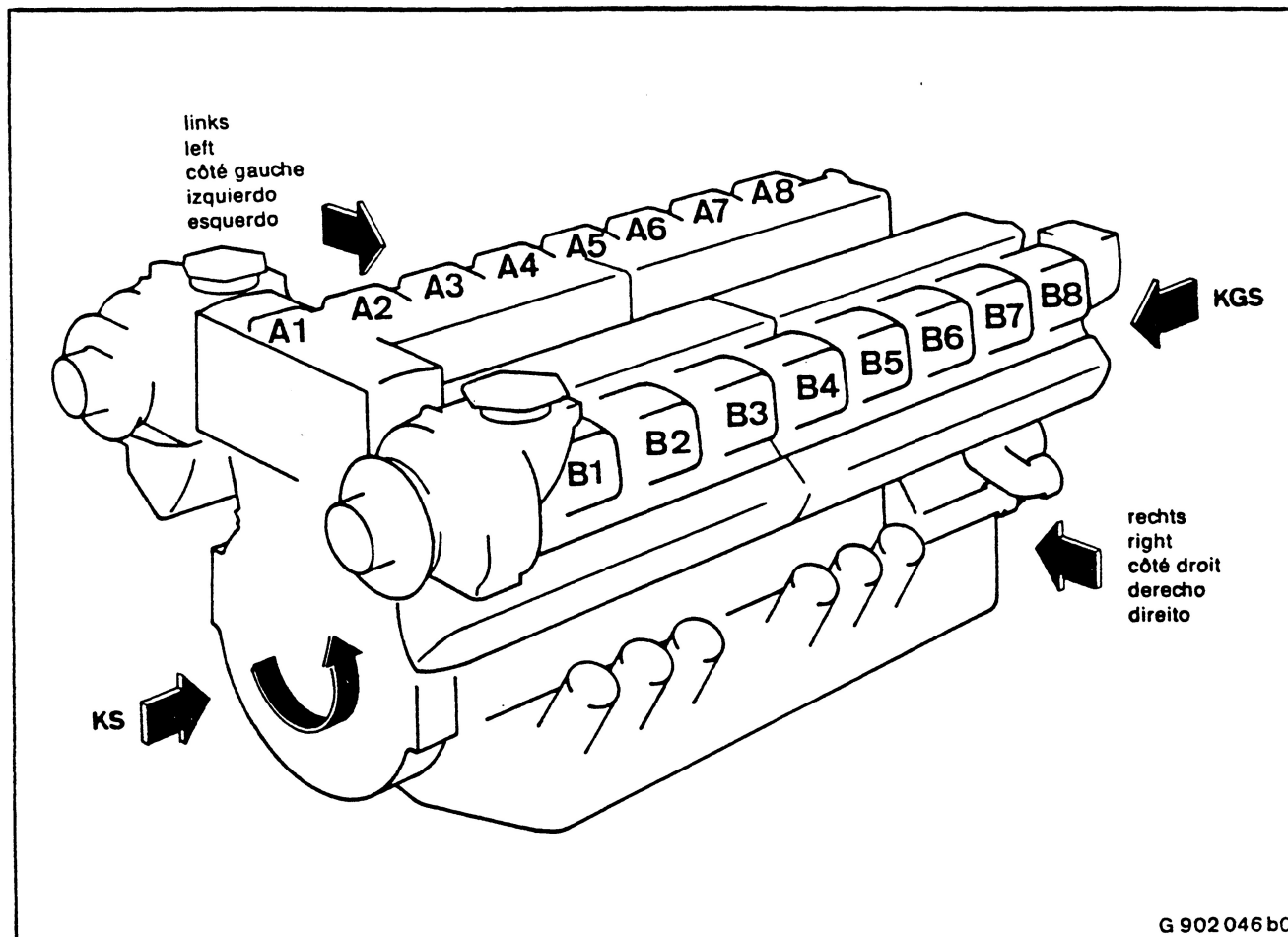
Engine Layout Cutaway Drawing 16V 396 TE.4 (also applies to 8 and 12V)



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- 1 Coolant expansion tank
- 2 Coolant thermostat
- 3 Fuel injection pump
- 4 Valve gear
- 5 Exhaust outlet
- 6 Air inlet
- 7 Emergency air shut-off flap
- 8 Intercooler
- 9 Exhaust turbocharger
- 10 Exhaust manifold
- 11 Engine mount, flywheel end

- 12 Starter
- 13 Cylinder head
- 14 Oil pan
- 15 Crankcase
- 16 Running gear
- 17 Engine oil pump
- 18 Vibration damper
- 19 Engine mount, timing end
- 20 Bilge pump
- 21 Fuel injection timer
- 22 Coolant cooler

ENGINE SIDE AND CYLINDER DESIGNATIONS

Engine sides are always designated as viewed from the flywheel end.

The left bank of cylinders is marked "A" and the right bank "B" (according to DIN standard 6265).

The cylinders of each bank are numbered consecutively, starting with No. 1 at the flywheel end.

Numbering of other engine components is also from the flywheel end, starting with No. 1.

Designations and abbreviations

Flywheel end = KS (formerly HKS)

Timing end = KGS (formerly GKS)

Left side (formerly A side)

Right side (formerly B side)

GENERAL SPECIFICATIONS

Operating method	four-stroke cycle, single-acting
Combustion method	direct injection
Mode of supercharging	single-stage sequential exhaust turbocharging
Cooling method	liquid cooling
Cylinder arrangement	90° Vee
Bore	165 mm
Stroke	185 mm
Displacement, cylinder	3.96 liters
Number of cylinders	16
Displacement, total	63.3 liters
Compression ratio	12.3 : 1
Direction of rotation (viewed from flywheel end)	c.c.w. (non-reversible)
Firing order	A1 A3-A5 A2-A8 A7 A4-A6 B5 B2-B8 B3 B4-B6 B1-B7
Fuel injection pressure:	
Setting pressure with new spring	280 to 288 bar
Specified pressure with run-in spring	270 to 278 bar
Minimum pressure with run-in spring	260 bar
Final compression pressure at 120 rpm, 1000 mbar and engine coolant at 5°C	20 to 24 bar
Break-away torque at engine coolant temperature of 40°C, allowance for acceleration included	approx. 800 Nm
Cranking torque at engine coolant temperature of 40°C	approx. 750 Nm
Firing speed at engine coolant temperature of 40°C	approx. 120 rpm
Mean piston speed	at 1900 rpm = 11.72 m/s
Engine noise	approx. 105 dB(A)
Mean free-field level	1 m distance
Exhaust noise, undamped	approx. 100 dB(A)
Free-field level	1 m distance